

provides global coverage—covering even the remote areas without compromising on the quality of signal. It can provide internet access to air space, in flight as well as on board ships and liners.

The system also provides a better security system, being a private network. An efficient encrypted satellite network is more secure than the terrestrial networks used by governments, defence, and other departments.

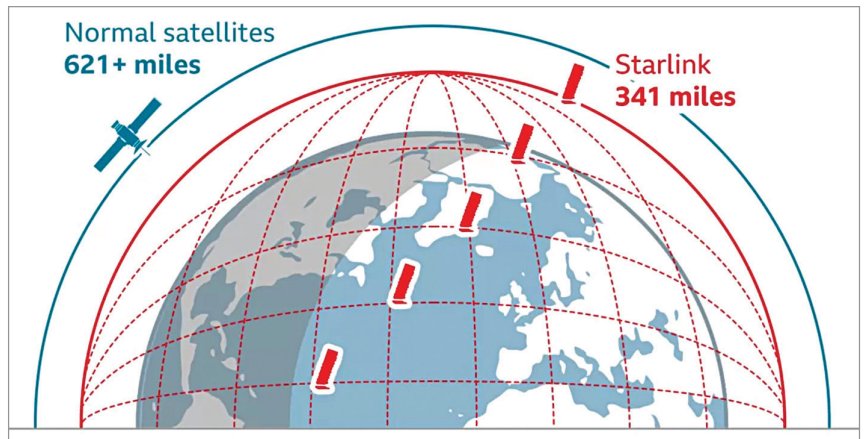
While terrestrial network systems involve numerous on-ground equipment prone to damage in cases of any disruptions or disasters, satellite internet is more resilient given its much less on-ground equipment. The satellite based internet service proves to be more cost-effective when used on a large scale, since it has lower set-up cost than the terrestrial networks that involve higher investments.

Disadvantages of satellite internet

While satellite internet has certain advantages, it also comes with certain drawbacks.

Limited speed is a major disadvantage since, as per the current technology, the speed is constrained and lies in the range of 50 to 150Mbps.

Weather related disturbances affect the signal path of satellite



Starlink satellites operate in low orbit (Source: <https://www.bbc.com>)

networks more vis-à-vis terrestrial networks. In case of rain, storms, and strong winds, the signal could get deteriorated with the chances of the internet being completely interrupted.

Also, high winds could cause physical damage to the dish. Satellite signals are more likely to get attenuated due to obstructions such as tree branches and buildings that come in the path between the dish antenna and the satellite.

The worldwide competition

Several companies that were providing internet connections using geosynchronous satellites started going bankrupt burdened by the expenses involved in launching the satellites and operating and maintaining the systems. Complex electronics was

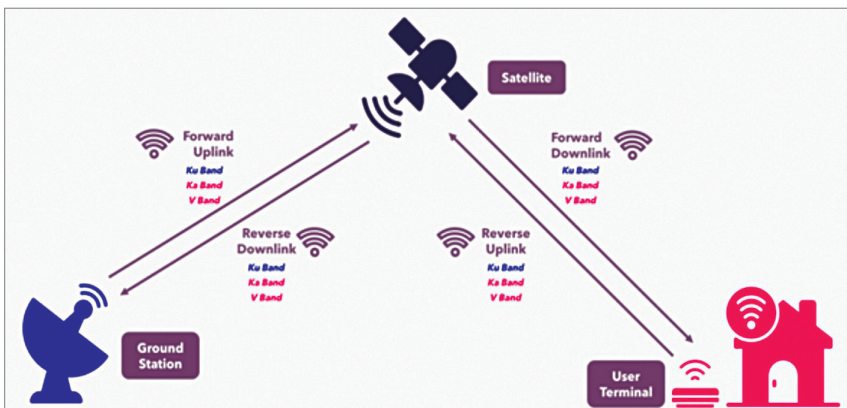
involved in the systems and there was a slump in market demand due to limited bandwidth, high latency, and prices.

It was in the 2010s that the organisations like Starlink, OneWeb, SpaceX, Amazon, Samsung, Boeing, Roscosmos, and Hongwan started developing internet services using LEO satellites. Starlink was the first to launch LEO satellites in 2019 and provide beta services in 2020. Later, OneWeb, SES, Eutelsat, and Hispasat followed their footsteps.

SpaceX Starlink

SpaceX started the project Starlink in 2015 in the USA with the aim of making available satellite broadband internet services using a network of thousands of interlinked desk-sized satellites in low Earth orbit about 550km above Earth. Each satellite weighs about 250kg and SpaceX plans to deploy 42,000 satellites over a period to provide worldwide internet coverage using the Ka and Ku frequency bands.

According to Jonathan McDowell, an astrophysicist and expert tracker of spaceflight activity, currently 3,858 Starlink satellites are in orbit making available services in 32 countries. Starlink is experimenting with lasers to enable satellites to talk to each other directly at the speed of light (faster than the fibre-



How Starlink works (Source: <https://www.clevernotes.in>)